ECE 329
Lecture 9

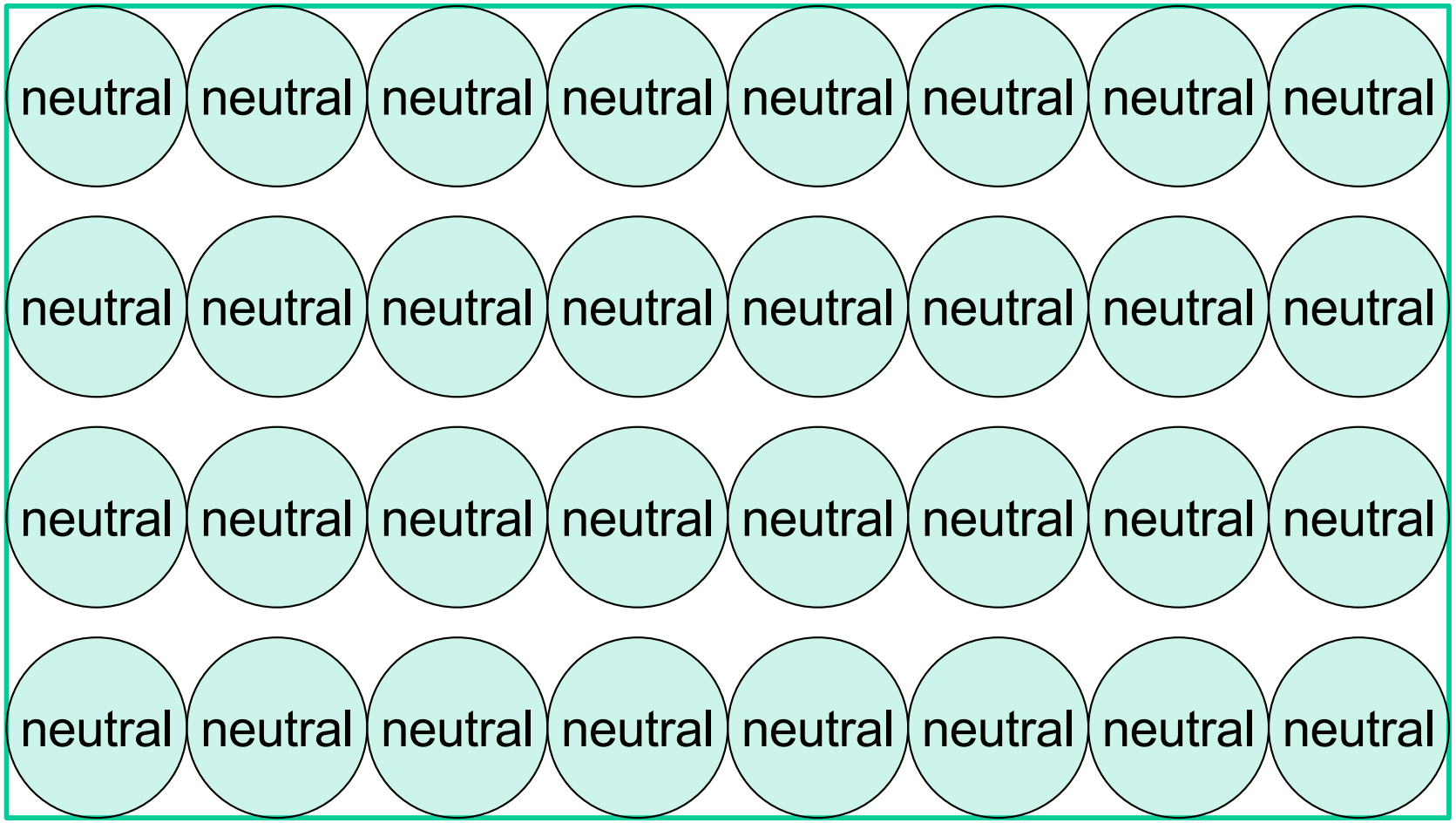
Dielectrics

Dr. Victoria Shao
yangshao@Illinois.edu

© Lynford Goddard, Brian Cunningham, Victoria Shao
- Copying, publishing or distributing without express written permission is prohibited.

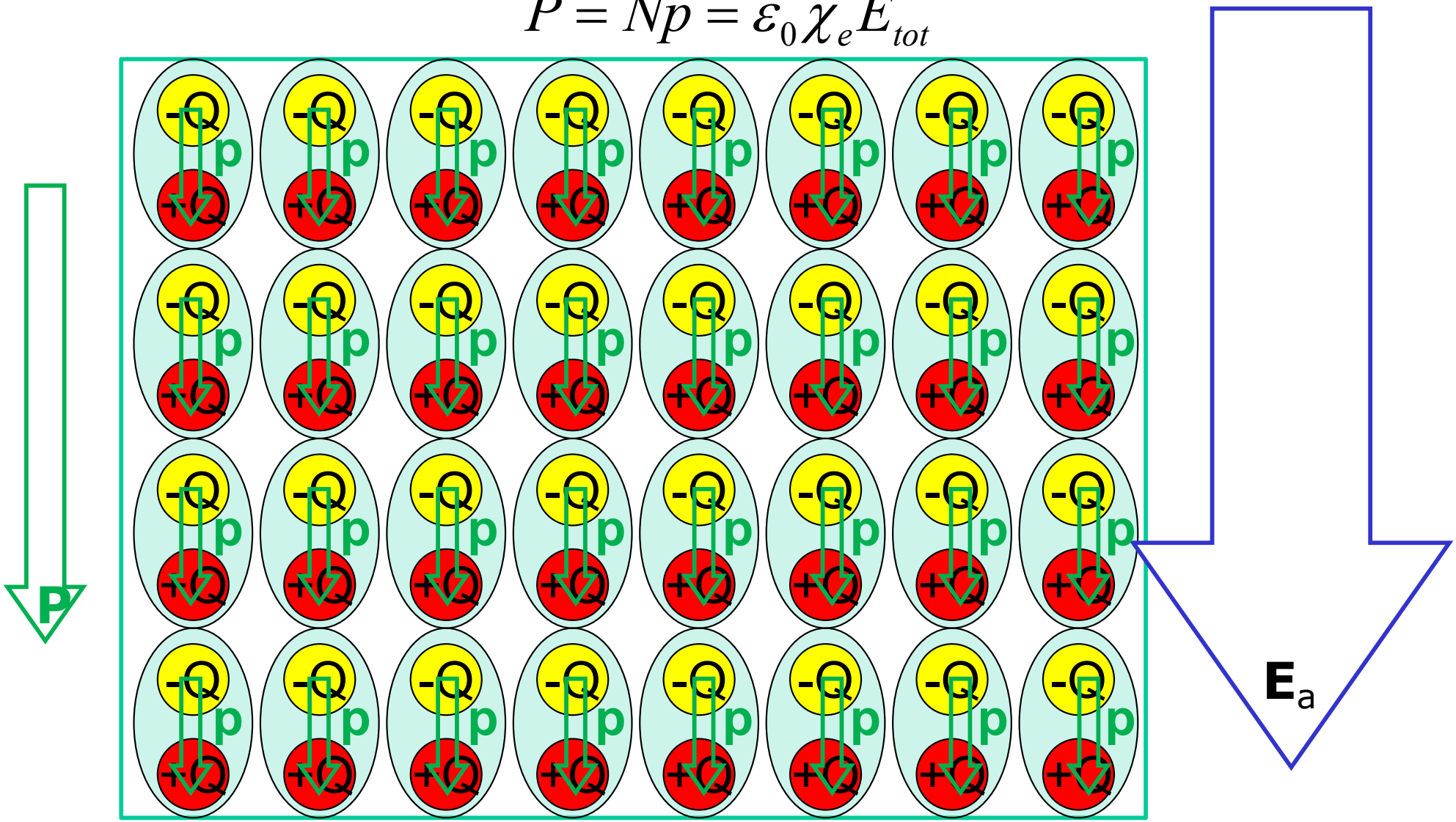
Apply **E** to Dielectric Material

Cancellation of internal charges



Apply **E** to Dielectric Material

$$\vec{P} = N\vec{p} = \varepsilon_0 \chi_e \vec{E}_{tot}$$



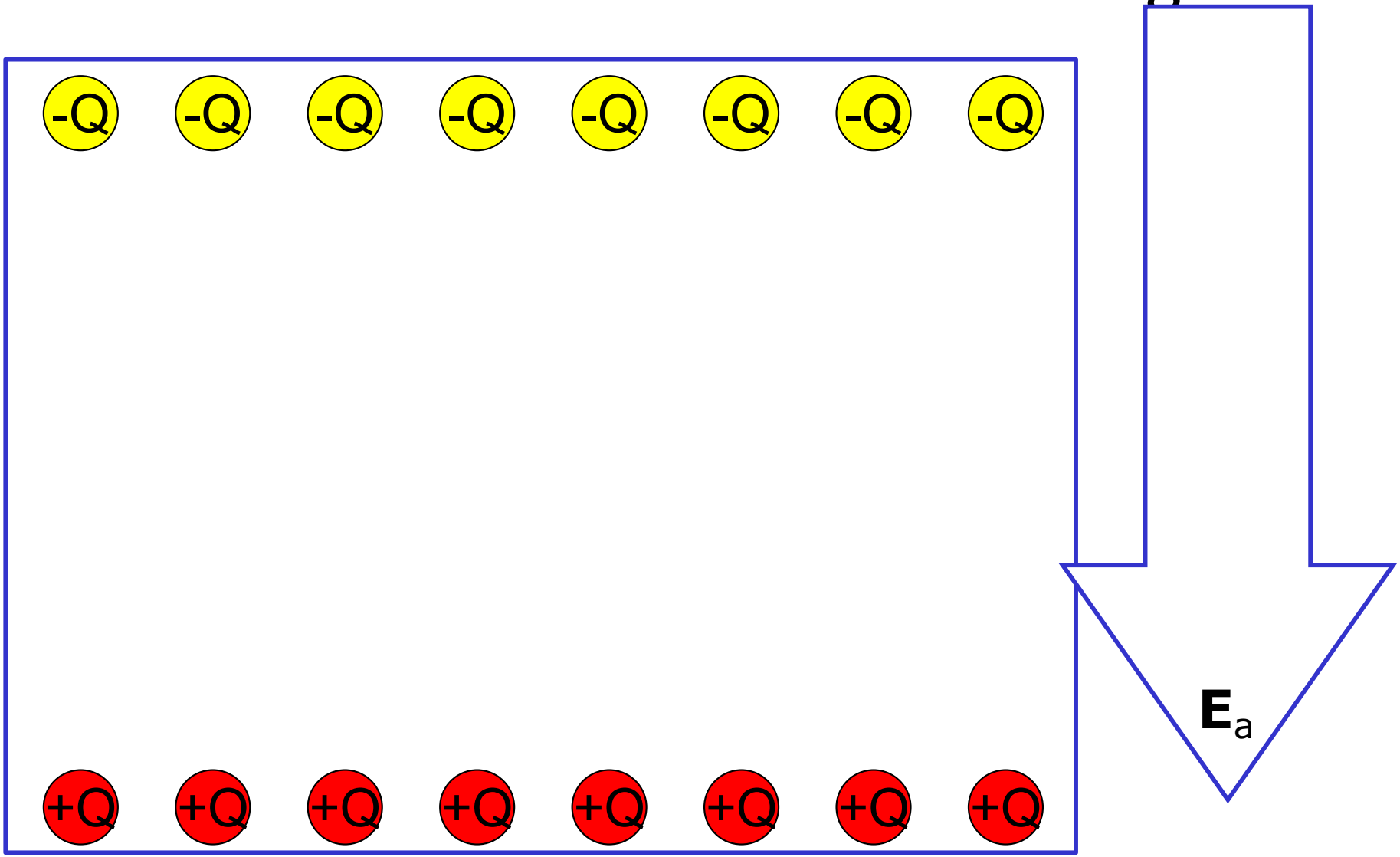
Dielectric Susceptibility χ_e

- Measures how easy it is to shift electrons from their centered orbit around the nuclei of a material to form internal dipoles

$$\vec{P} = \varepsilon_0 \chi_e \vec{E}_{tot}$$

Apply **E** to Dielectric Material

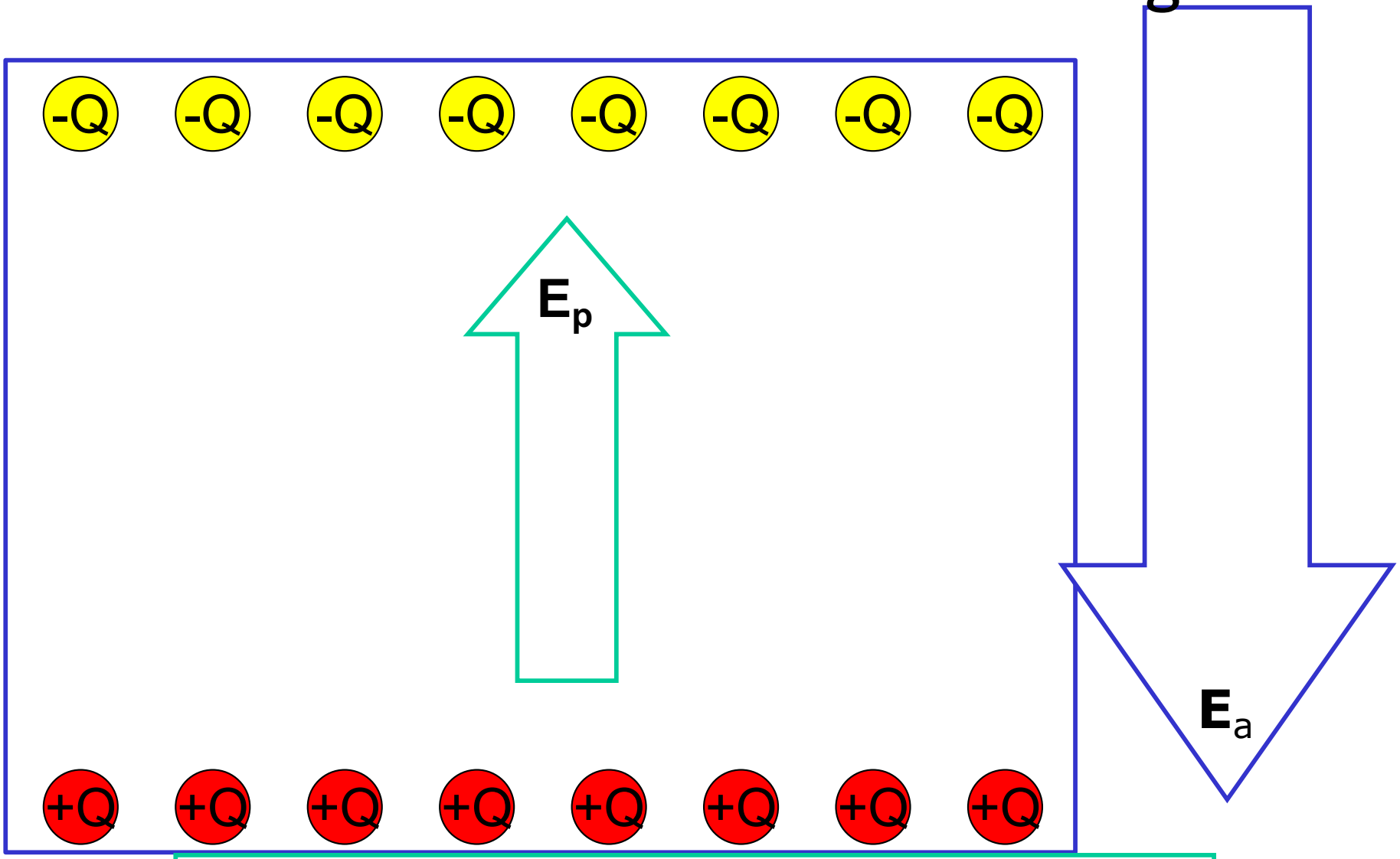
Cancellation of internal charges



"Polarization Surface Charges" Remain

Apply **E** to Dielectric Material

Cancellation of internal charges



Secondary Electric Field Produced

* *Does not cancel applied field inside dielectric*

$\vec{E}_{tot}$ is reduced but not zero

Unlike conductors where **E**=0, in the dielectric slab, the total field is **reduced (but NOT eliminated)** by the secondary field produced by the surface polarization charge

$$\vec{E}_{tot} = \vec{E}_a + \vec{E}_p$$

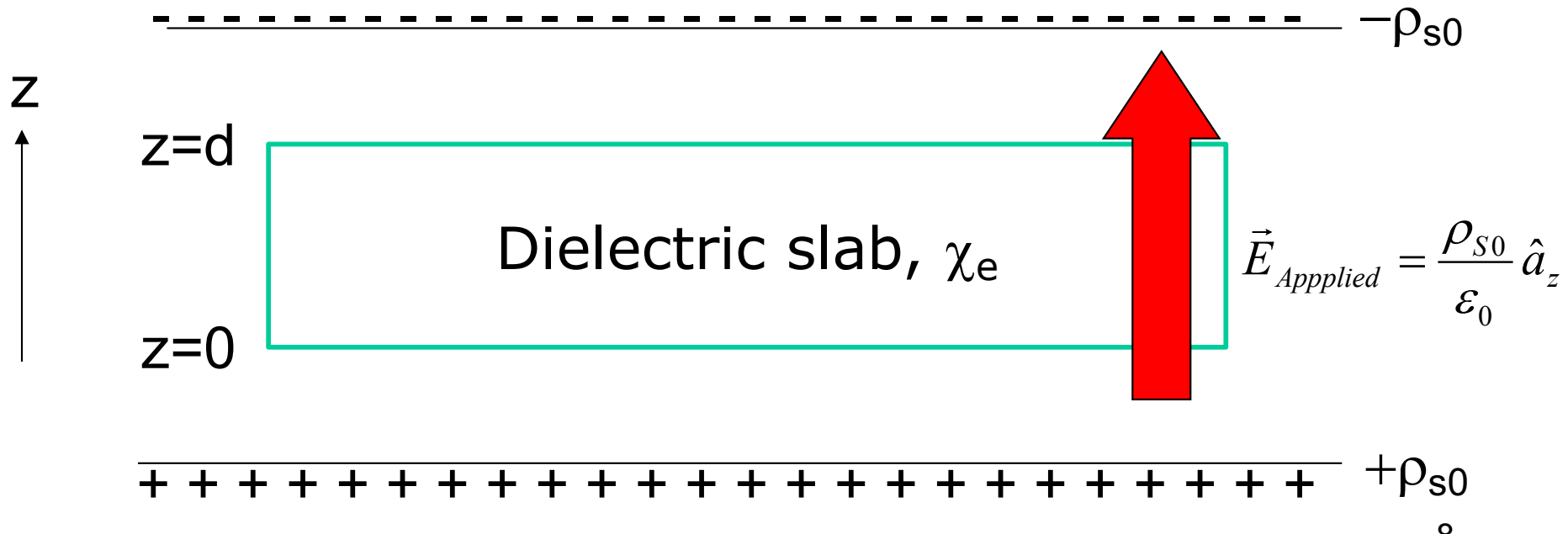
And the polarization vector, $\vec{P}$, is the polarization of atoms in the dielectric due to the TOTAL $\vec{E}_a$ field

$$\vec{P} = \epsilon_0 \chi_e \vec{E}_{tot}$$

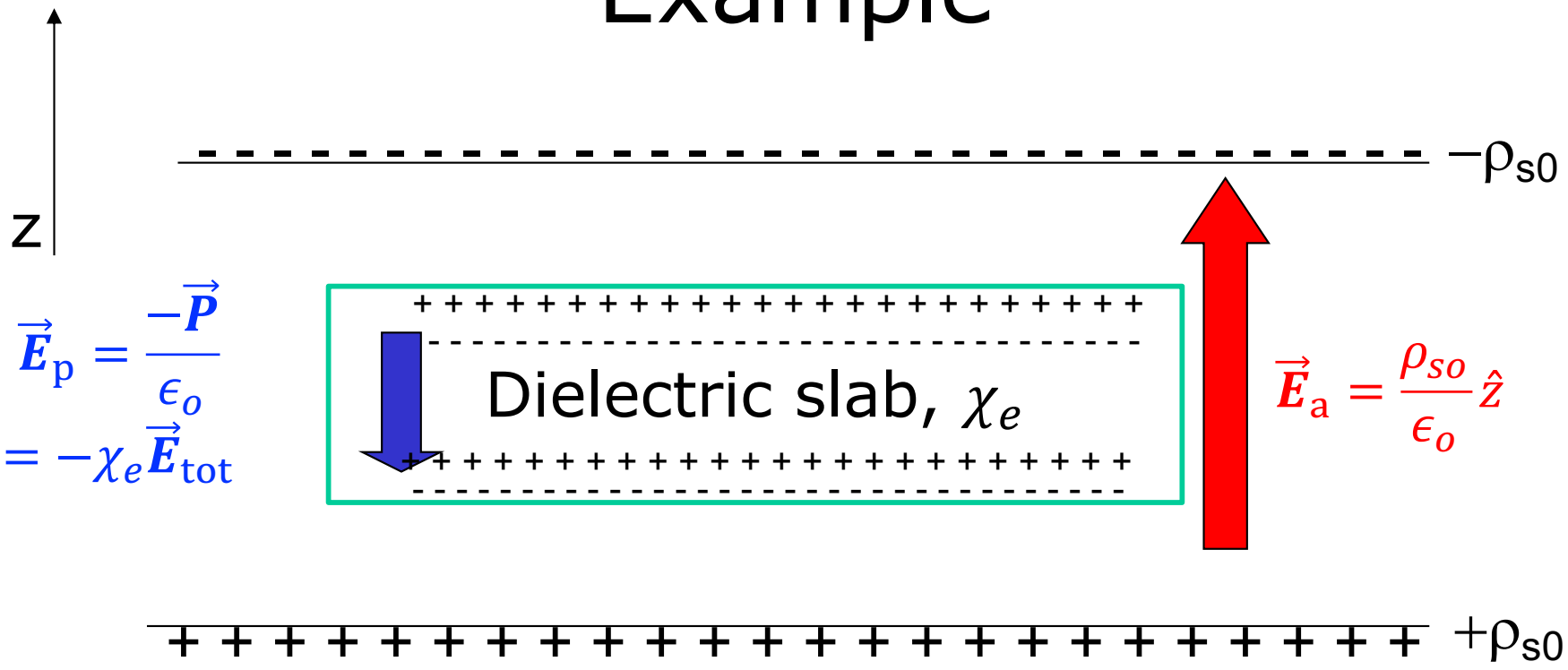
Electric susceptibility χ_e is a proportional constant.

Example

- Apply external electric field to a slab of dielectric material
- Apply field by placing the slab between two equal and opposite charge densities



Example



$$\vec{E}_{tot} = \vec{E}_a + \vec{E}_p = \vec{E}_a - \chi_e \vec{E}_{tot}$$

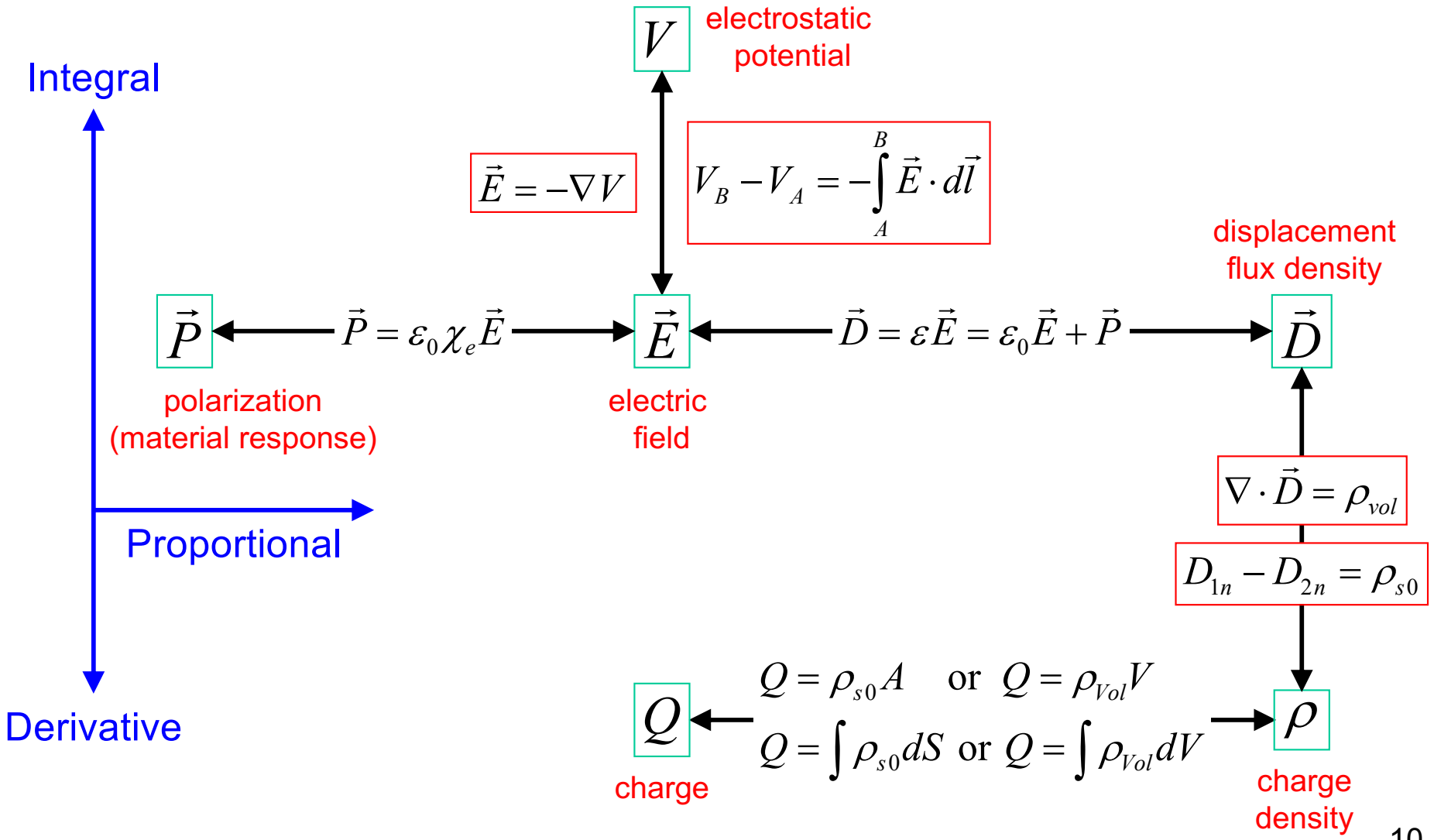
$$\vec{E}_{tot} = \frac{\vec{E}_a}{1+\chi_e}$$

E-field strength reduced by $(1+\chi_e)$

$$\vec{D} \equiv \epsilon \vec{E}_{total} = \epsilon_0 \vec{E}_{total} + \vec{P}$$
$$= \epsilon_0 (1+\chi_e)\vec{E}_{total} = \rho_{so}\hat{z}$$

D-field strength is same as free space because charge is fixed

Connection of Concepts for Electrostatics



Example

So finally, the “final answer” is:

$$\vec{\boldsymbol{E}}_{tot} = \frac{\vec{\boldsymbol{E}}_a}{1+\chi_e} = \frac{\rho_{so}/\epsilon_o}{1+\chi_e}$$

The total electric field strength inside the dielectric is reduced from its “free space” value by $(1+\chi_e)$

Free space: $\chi_e=0$ Perfect Conductor: $\chi_e \rightarrow \infty$

Gauss's law

In free space, Gauss's law is

$$\nabla \cdot \vec{\boldsymbol{D}} = \nabla \cdot \epsilon_o \vec{\boldsymbol{E}} = \rho_f \Longrightarrow$$ free charge

In material medium, total charge becomes

$$\rho = \rho_f + \rho_b \Longrightarrow$$ free charge + bond charge

$$\nabla \cdot \vec{\boldsymbol{D}} = \nabla \cdot \left(\epsilon_o \vec{\boldsymbol{E}} + \vec{\boldsymbol{P}}\right) = \rho_f$$

$$\boxed{\vec{\boldsymbol{D}} = \epsilon_o \vec{\boldsymbol{E}} + \vec{\boldsymbol{P}}}$$

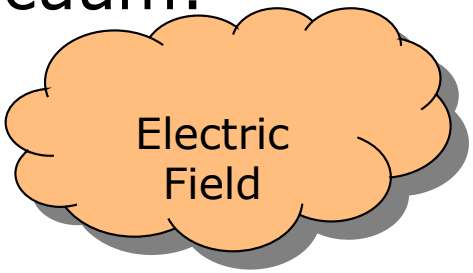
or $$\nabla \cdot \epsilon_o \vec{\boldsymbol{E}} = \rho_f - \nabla \cdot \vec{\boldsymbol{P}}$$

here $$\rho_b = -\nabla \cdot \vec{\boldsymbol{P}}$$ bound charge density

$$\boxed{\nabla \cdot \vec{\boldsymbol{D}} = \rho}$$ Gauss's Law regardless of material medium

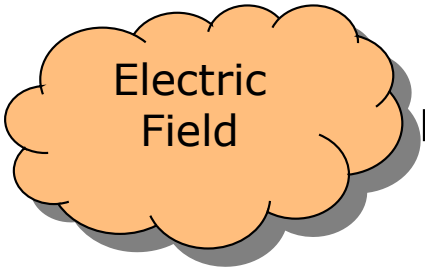
Displacement field $\vec{D}$

In vacuum:



Displacement
(electric flux density)
$\vec{D} = \epsilon_o \vec{\mathbf{E}}$

In dielectric:



Displacement
(electric flux density)
$\vec{D} = \epsilon_o \vec{\mathbf{E}} + \vec{P}$

Item $\epsilon_o \vec{\mathbf{E}}$: is the same as the case in vacuum

Item "$\vec{P} = \epsilon_0 \chi_e \vec{E}_{tot}$": is the polarization due to contribution of dipole formation in dielectric under applied $\vec{\mathbf{E}}$ field

Dielectric Constant ϵ

$$\vec{\boldsymbol{D}} = \vec{\boldsymbol{P}} + \epsilon_0 \vec{\boldsymbol{E}} \qquad \vec{\boldsymbol{P}} = \epsilon_0 \chi_e \vec{\boldsymbol{E}}$$

$$\vec{\boldsymbol{D}} = \epsilon_0 \chi_e \vec{\boldsymbol{E}} + \epsilon_0 \vec{\boldsymbol{E}}$$

$$\vec{\boldsymbol{D}} = \epsilon_0 (1 + \chi_e) \vec{\boldsymbol{E}}$$

$$\boxed{\vec{\boldsymbol{D}} = \epsilon_0 \epsilon_r \vec{\boldsymbol{E}}}$$

$$\boxed{\begin{aligned} \varepsilon_r &= (1 + \chi_e) \\ \varepsilon &= \varepsilon_0 \varepsilon_r \end{aligned}}$$

ϵ_r: Relative Permittivity or "Dielectric Constant"

Dielectric Permittivity of the Material ϵ [F/m]

$$\boxed{\vec{\boldsymbol{D}} = \epsilon \vec{\boldsymbol{E}}}$$

Units for ε_r: None

Units for χ_e: None

Infinite plane dielectric slab

- An infinite plane dielectric slab lies between two infinite plane sheets of uniform charge density of $\rho = \pm 1 \mu C/m^2$. Find **D**, **E**, and **P** inside the slab.



From D4.3 (p 226) of old book

Challenge Question: Spherical shell dielectric

- If we have:

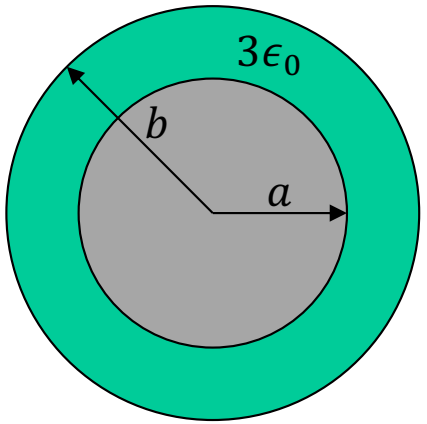
$$\begin{cases} r < a & perfect\ conductor \\ a < r < b & perfect\ dielectric\ \varepsilon = 3\varepsilon_0 \\ r > b & perfect\ conductor \end{cases}$$

with equally and oppositely charged PECs and **D** points radially inward, which is true:

(a) V(b) > V(a)

(b) V(b) = V(a)

(c) V(b) < V(a)



Lecture 9 Summary

- Polarization

$$\vec{\boldsymbol{P}} = \epsilon_0 \chi_e \vec{\boldsymbol{E}}_{tot} = \epsilon_0 \chi_e (\vec{\boldsymbol{E}}_a + \vec{\boldsymbol{E}}_p)$$

  - Homogenous dielectric medium
    - Reduces **E**-field strength by $(1+\chi_e)$
    - $\vec{\boldsymbol{D}}$ has same value as free space
    - $\vec{\boldsymbol{D}}$, $\vec{\boldsymbol{E}}$, and $\vec{\boldsymbol{P}}$ point in the same direction
  - New definition $\vec{\boldsymbol{D}} = \vec{\boldsymbol{P}} + \epsilon_0 \vec{\boldsymbol{E}}$
- Next class
  - Capacitance and Conductance